

Control Barrier Functions

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Introduction

Control Barrier Functions (CBF) are a methodology for creating “safe zones” or “danger zones” in control systems. Our proposal is to explore applications of CBFs on a variety of reinforcement learning systems. We plan to leverage Gymnasium simulations to practice our training loop and testing evaluations. Each team member will conduct exploratory analysis on a different autonomous, robotic or mechanical system.

Resources, Code, and Datasets

- Ames, Aaron D., et al. "Control barrier functions: Theory and applications." 2019 18th European control conference (ECC). Ieee, 2019.
- https://bolundai0216.github.io/assets/images/control_theory/An%20Introduction%20to%20Control%20Barrier%20Functions.pdf
- Towers, Mark, et al. "Gymnasium: A standard interface for reinforcement learning environments." arXiv preprint arXiv:2407.17032 (2024).
- MIT Lincoln Laboratory Terminal Encounter Model (LLTEM) V1.0 Dataset: Will be used to generate realistic trajectories to test the DAA-specific safety barrier.
- RTCA, "Minimum Operational Performance Standards (MOPS) for Detect and Avoid (DAA) Systems," DO-365, 2017.

Team Contribution

Each member will choose their domain and collaborate with teammates about how to apply algorithms found in the literature to these simulated datasets. Some possible domains are quadrotor dynamics, bipedal walking, robotic pick-and-place, and satellite attitude control. It will be up to the individual to determine what bounds are appropriate for the “unsafe” zone to be avoided in the trajectory planning phase.

Association with Classwork

This project topic is an extension of the reinforcement learning procedures learned in ME 569. The paper by Choi et al (2025) also talks about optimal placement of sensors in a dynamical system, which could be applicable to ME 570. We will use SINDy (ME 571) to extract the governing equations of a dynamical system from noisy observations, which then serves as the foundation for the Control Barrier Functions.